

# The T0 Energy Field Model: Mathematical Formulation

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### Correction notes

**Corrections from the calculation review (30 Sept. 2026).** The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- $E_0$ : The two values are the same  $E_0$ , bare ( $\sqrt{m_e m_\mu} = 7.348$  MeV) and fractally corrected ( $E_0/\sqrt{K_{\text{frak}}} = 7.398$  MeV, Doc. 190, R72). The corrected value serves as the anchor for  $\alpha_i$  from the masses alone,  $\alpha = 1/137.059$  (+0.017 %).
- $\alpha$ : agreement 0.0005 % instead of 0.03 %.
- Anomalous magnetic moments: the short formula  $a_\ell = \frac{\xi}{2\pi}(E_\ell/E_e)^2$  has been removed; recalculated it gave  $a_\mu = 0.907$  instead of  $1.166 \times 10^{-3}$ . Authoritative are Docs. 018 and 158.
- Dimension of the Lagrangian density:  $[E^4]$ , not  $[E^2]$ .
- $G_{\text{SI}}$  formula:  $\frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \rightarrow \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}$  – with  $C_{\text{dim}}$  but without the compensating factor  $E_{\text{char}}$ ,  $G$  would be too small by a factor of  $E_{\text{char}} = 28.8$ . Likewise:  $\xi^2/(4m_e)$  has  $[E^{-1}]$ , not  $[E^{-2}]$ .
- Lepton mass table: 0.507/103.5/1815  $\rightarrow$  0.505/104.96/1783.4 MeV (corpus ladder).

## Abstract

The T0 model describes physical phenomena through a universal energy field  $E_{\text{field}}(x, t)$  with the parameter  $\xi = \frac{4}{3} \times 10^{-4}$ . The field equation is  $\square E_{\text{field}} = 0$ , the Lagrangian density  $\mathcal{L} = \xi(\partial E)^2$ . The model uses standard natural units with  $\hbar = c = 1$ .

### Fundamental quantities:

- Characteristic energy:  $E_0 = \sqrt{m_e \cdot m_\mu} = 7.348 \text{ MeV (bare)}$ ; fractally corrected  $E_0/\sqrt{K_{\text{frak}}} = 7.398 \text{ MeV}$  (anchor for  $\alpha$ )
- Fine structure constant:  $\alpha = \xi(E_0/1 \text{ MeV})^2 \approx 1/137$
- Gravitational constant:  $G = \xi^2/(4m_e) \times \text{factors}$

**Predictions:** Lepton masses with 2% accuracy, fine structure constant  $\alpha \approx 1/137$  (from the masses alone  $1/137.059, +0.017 \%$ ).

**Detailed derivations:** See Document 011 (fine structure), 012 (gravitation), 018 (g-2 geometric), 019 (Lagrangian).

## .1 Units Convention

### T0 natural units (Heaviside-Lorentz)

T0 uses Heaviside-Lorentz natural units:

$$\hbar = c = 4\pi\epsilon_0 = 1 \quad (1)$$

In this system:

- Fine structure constant:  $\alpha = e^2 = 1$  (by convention in T0)
- Energy = Mass:  $E = m$
- Length = Time = Energy<sup>-1</sup>:  $[L] = [T] = [E^{-1}]$

**Note:** In SI units,  $\alpha \approx 1/137$ . The choice  $\alpha = 1$  in T0 is a unit convention that simplifies formulas.

### Dimensions in natural units

$$[E] = E \quad (2)$$

$$[m] = E \quad (3)$$

$$[t] = E^{-1} \quad (4)$$

$$[L] = E^{-1} \quad (5)$$

$$[G] = E^{-2} \quad (6)$$

$$[\partial_\mu] = E \quad (7)$$

## .2 Time-Energy Duality

### Fundamental relation

$$T_{\text{field}}(x, t) \cdot E_{\text{field}}(x, t) = 1 \quad (8)$$

with  $[T_{\text{field}}] = E^{-1}$  and  $[E_{\text{field}}] = E$ .

## Intrinsic time field

$$T_{\text{field}}(x, t) = \frac{1}{E_{\text{field}}(x, t)} \quad (9)$$

## .3 Universal Field Equation

### Wave equation

$$\square E_{\text{field}} = 0 \quad (10)$$

with d'Alembert operator:

$$\square = \nabla^2 - \frac{\partial^2}{\partial t^2} \quad (11)$$

### With sources

$$\nabla^2 E_{\text{field}} = 4\pi G\rho \cdot E_{\text{field}} \quad (12)$$

Dimension check:  $[E^3] = [E^{-2}][E^4][E] = [E^3] \checkmark$

## .4 Lagrangian Density

### Universal Lagrangian density

$$\mathcal{L} = \xi \cdot (\partial_\mu E_{\text{field}})(\partial^\mu E_{\text{field}}) \quad (13)$$

with  $\xi = \frac{4}{3} \times 10^{-4}$ .

### Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial E} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu E)} = 0 \quad (14)$$

yields:

$$\square E_{\text{field}} = 0 \quad (15)$$

## .5 Characteristic Energy

### Definition

The characteristic energy  $E_0$  is the geometric mean of electron and muon mass (derivation in Document 011):

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (16)$$

## Numerical values

From experimental masses:

$$E_0 = \sqrt{0.511 \times 105.66} \quad (17)$$

$$= \sqrt{53.99} \quad (18)$$

$$= 7.348 \text{ MeV} \quad (19)$$

Fractally corrected value (Doc. 190, R72):

$$E_0^{T0} = \frac{E_0}{\sqrt{K_{\text{frak}}}} = \frac{7.348}{\sqrt{1 - 100\xi}} = 7.397 \approx 7.398 \text{ MeV} \quad (20)$$

Both values are the same  $E_0$ , bare and fractally corrected: absolute values carry  $K_{\text{frak}}$ , ratios do not. The square-root factor follows because  $E_0$  enters  $\alpha = \xi E_0^2$  quadratically. The corrected value 7.398 MeV serves as the anchor for  $\alpha$ .

(Corrected on 30 Sept. 2026: previously "Theoretical T0 value:  $E_0^{T0} = 7.398 \text{ MeV}$ ; Deviation: 0.7 % (within geometric corrections)". The relation  $7.398/7.348 = 1.00682 \approx 1/\sqrt{K_{\text{frak}}} = 1.00673$  (0.008 %) was not stated.)

## Usage

$E_0$  serves as energy scale for:

- Fine structure constant:  $\alpha = \xi(E_0/1 \text{ MeV})^2$
- Normalization of electromagnetic effects

## .6 The Parameter $\xi$

### Definition

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333 \times 10^{-4} \quad (21)$$

Dimensionless:  $[\xi] = 1$ .

### Geometric components

$$\xi = G_3 \times S_{\text{ratio}} \quad (22)$$

where:

- $G_3 = \frac{4}{3}$ : Geometric factor (sphere-cube ratio)
- $S_{\text{ratio}} = 10^{-4}$ : Scale ratio

## .7 Fine Structure Constant

### In T0 units

In T0 natural units:  $\alpha = 1$  (by convention)

## Reconstruction of SI value

**Important:** The SI value can be reconstructed from T0 parameters!

$$\alpha_{\text{SI}} = \xi \cdot \left( \frac{E_0}{1 \text{ MeV}} \right)^2 \quad (23)$$

## Numerical calculation

With  $\xi = \frac{4}{3} \times 10^{-4}$  and  $E_0 = 7.398 \text{ MeV}$ :

$$\alpha_{\text{SI}} = 1.3333 \times 10^{-4} \times (7.398)^2 \quad (24)$$

$$= 1.3333 \times 10^{-4} \times 54.73 \quad (25)$$

$$= 7.297 \times 10^{-3} \quad (26)$$

$$= \frac{1}{137.04} \quad (27)$$

Experimental:  $\alpha_{\text{exp}} = \frac{1}{137.036}$

Agreement: 0.0005 % (with the anchor  $E_0 = 7.398 \text{ MeV}$ ). From the masses alone,  $\alpha = \xi E_0^2 / K_{\text{frak}}$  with  $E_0 = 7.348 \text{ MeV}$  gives  $\alpha = 1/137.059$  (+0.017 %).

(Corrected on 30 Sept. 2026: previously "Agreement: 0.03%";  $1/137.035$  against  $1/137.036$  is 0.0005 %.)

## Dimension check

$$[\alpha_{\text{SI}}] = [\xi] \times \left[ \frac{E}{E} \right]^2 = 1 \times 1 = 1 \quad \checkmark \quad (28)$$

# .8 Gravitational Constant

## T0 formula

The gravitational constant is derived from  $\xi$  and  $m_e$  (derivation in Document 012):

$$G = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (29)$$

(Corrected on 30 Sept. 2026: previously  $\frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} - C_{\text{dim}} = 1/E_{\text{char}} = 3.472 \times 10^{-2}$  is cancelled again by the factor  $E_{\text{char}} = 28.8$  in the step chain of Doc. 012; in the formula alone the product would give  $2.35 \times 10^{-12}$  instead of  $6.674 \times 10^{-11}$ .)

where:

- $C_{\text{dim}}$ : Dimension correction
- $C_{\text{conv}}$ : SI conversion factor

## Fundamental relation

In natural units:

$$\xi = 2\sqrt{G \cdot m_e} \quad (30)$$

Solved for  $G$ :

$$G_{\text{nat}} = \frac{\xi^2}{4m_e} \quad (31)$$

Dimension:  $[G] = [E^{-2}]$  in natural units; the right-hand side  $\xi^2/(4m_e)$ , however, has  $[E^{-1}]$  – the missing factor  $1/E_{\text{char}}$  and the SI conversion reside in  $C_{\text{conv}}$  (Doc. 012). *(Corrected on 30 Sept. 2026: previously without this remark, so that  $\xi^2/(4m_e)$  appeared as  $[E^{-2}]$ .)*

## .9 Characteristic Lengths

### T0 characteristic length

$$r_0 = 2GE \quad (32)$$

Dimension:  $[r_0] = [E^{-2}][E] = [E^{-1}] = [L] \checkmark$

### Derivation

For spherically symmetric point source  $\rho(r) = E_0 \delta^3(\vec{r})$ :

Solution of  $\nabla^2 E = 4\pi G \rho E$ :

$$E(r) = E_0 \left(1 - \frac{r_0}{r}\right) \quad (33)$$

with  $r_0 = 2GE_0$ .

### Time scale

$$t_0 = \frac{r_0}{c} = r_0 = 2GE \quad (34)$$

(since  $c = 1$ )

## .10 Scale Hierarchy

### Planck length as reference

$$\ell_P = \sqrt{G} = 1 \quad (\text{in nat. units}) \quad (35)$$

### Scale ratio

$$\xi_{\text{ratio}} = \frac{\ell_P}{r_0} = \frac{\sqrt{G}}{2GE} = \frac{1}{2\sqrt{G} \cdot E} \quad (36)$$

For  $E \sim 1 \text{ GeV}$ :

$$\frac{r_0}{\ell_P} = 2\sqrt{G}E = \frac{2E}{E_P} \sim 1.6 \times 10^{-19} \quad (\text{sub-Planck}) \quad (37)$$

*(Corrected on 29 Sept. 2026: previously  $r_0/\ell_P \sim 10^7$  – with  $r_0 = 2GE$  and  $\ell_P = \sqrt{G}$  one has  $r_0/\ell_P = 2E/E_P$ , i.e.  $1.6 \times 10^{-19}$  for  $E = 1 \text{ GeV}$ .)*

| Particle | Energy [MeV] |
|----------|--------------|
| Electron | 0.511        |
| Muon     | 105.658      |
| Tau      | 1776.86      |

## .11 Particles as Field Excitations

### Classification by energy

#### Antiparticles

Negative field excitations:  $E_{\text{field}} < 0$

## .12 Lepton Masses

The T0 model predicts lepton masses (derivation in Document 003):

| Lepton   | T0 [MeV] | Exp [MeV] | $\Delta$ [%] |
|----------|----------|-----------|--------------|
| Electron | 0.505    | 0.511     | -1.18        |
| Muon     | 104.96   | 105.66    | -0.66        |
| Tau      | 1783.4   | 1776.9    | +0.37        |

(Corrected on 30 Sept. 2026: previously 0.507/103.5/1815 MeV with 0.87/2.09/2.16 % – the corpus ladder  $m_i = r_i \xi^{p_i} v$  (Docs. 005, 006) gives 0.505/104.96/1783.4 MeV, i.e. -1.18/ -0.66/ +0.37 %; also  $(0.511 - 0.507)/0.511$  would be 0.78 %, not 0.87 %.)

## .13 Anomalous Magnetic Moments

Anomalous magnetic moments are treated in dedicated documents: Doc. 018 (with its revisions) and Doc. 158.

(Corrected on 30 Sept. 2026: previously this section gave the short formula  $a_\ell = \frac{\xi}{2\pi}(E_\ell/E_e)^2$  with  $a_\mu = 1.166 \times 10^{-3}$ ,  $a_e = 2.122 \times 10^{-5}$  and  $a_\tau = 1.28 \times 10^{-3}$ . Recalculated, the formula gives  $a_\mu = 2.122 \times 10^{-5} \times 42\,753 = 0.907$  and  $a_\tau = 256.6$ ;  $a_e$  lies a factor of 55 below the measured value  $1.15965 \times 10^{-3}$ . The formula has been removed, as have its repetitions in the dimensional analysis, formula reference and calculation examples.)

## .14 Three Field Geometries

### Type 1: Localized spherical

$$E(r) = E_0 \left(1 - \frac{\beta}{r}\right), \quad \beta = r_0 \quad (38)$$

Application: Individual particles (electron, muon, tau)

### Type 2: Localized non-spherical

$$E(\vec{r}) = E_0 \left(1 - \frac{\beta_{ij} r_i r_j}{r^3}\right) \quad (39)$$

Application: Bound systems

### Type 3: Extended homogeneous

Effective parameter:

$$\xi_{\text{eff}} = \frac{\xi}{2} = \frac{2}{3} \times 10^{-4} \quad (40)$$

Application: Cosmology (see Document 026)

## .15 Mathematical Identities

### Energy field normalization

$$E_{\text{field}}(\vec{r}, t) = E_0 \cdot f(\vec{r}, t) \cdot e^{i\phi(\vec{r}, t)} \quad (41)$$

with:

- $E_0$ : Characteristic energy
- $f(\vec{r}, t)$ : Normalized profile
- $\phi(\vec{r}, t)$ : Phase

### Duality consistency

Time-mass (Document 003):  $T \cdot m = 1$

Time-energy (this document):  $T \cdot E = 1$

In natural units ( $c = 1$ ):

$$E = mc^2 = m \quad \Rightarrow \quad T \cdot m = T \cdot E \quad (42)$$

## .16 Dimensional Analysis Verifications

### Field equation

$$[\nabla^2 E] = [L^{-2}][E] = [E^2][E] = [E^3] \quad (43)$$

$$[4\pi G \rho E] = [E^{-2}][E^4][E] = [E^3] \quad \checkmark \quad (44)$$

### Characteristic length

$$[r_0] = [2GE] = [E^{-2}][E] = [E^{-1}] = [L] \quad \checkmark \quad (45)$$

### Lagrangian density

$$[\mathcal{L}] = [\xi][(\partial E)^2] = [1][E^4] = [E^4] \quad (\text{correct for Lagrangian density}) \quad (46)$$

(Corrected on 30 Sept. 2026: previously  $[1][E^2] = [E^2]$ ;  $[\partial_\mu E] = E^2$ , hence  $[(\partial E)^2] = E^4$ , as in the symbol table.)

## .17 Formula Reference

### Fundamental equations

$$\text{Duality: } T_{\text{field}} \cdot E_{\text{field}} = 1 \quad (47)$$

$$\text{Wave equation: } \square E_{\text{field}} = 0 \quad (48)$$

$$\text{With sources: } \nabla^2 E = 4\pi G \rho E \quad (49)$$

$$\text{Lagrangian density: } \mathcal{L} = \xi(\partial E)^2 \quad (50)$$

### Derived constants

$$\text{Characteristic energy: } E_0 = \sqrt{m_e \cdot m_\mu} = 7.348 \text{ MeV (bare; corrected 7.398)} \quad (51)$$

$$\text{Fine structure constant: } \alpha = \xi(E_0/1 \text{ MeV})^2 \approx 1/137 \quad (52)$$

$$\text{Gravitational constant: } G = \frac{\xi^2}{4m_e} \times \text{factors} \quad (53)$$

### Characteristic scales

$$\text{T0 length: } r_0 = 2GE \quad (54)$$

$$\text{T0 time: } t_0 = 2GE \quad (55)$$

$$\text{Planck length: } \ell_P = \sqrt{G} = 1 \quad (56)$$

$$\text{Scale ratio: } \xi_{\text{ratio}} = \frac{1}{2\sqrt{GE}} \quad (57)$$

### Prediction formulas

$$\text{Parameter: } \xi = \frac{4}{3} \times 10^{-4} \quad (58)$$

$$\text{Effective parameter: } \xi_{\text{eff}} = \frac{\xi}{2} \quad (59)$$

## .18 Numerical Values

### Fundamental constants (in natural units)

$$\hbar = 1 \quad (60)$$

$$c = 1 \quad (61)$$

$$\alpha = \frac{1}{137.036} \approx 7.297 \times 10^{-3} \quad (62)$$

$$G = 1 \text{ (numerically, dimension } [E^{-2}]) \quad (63)$$

## T0 parameters

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333 \times 10^{-4} \quad (64)$$

$$\xi^2 = 1.7778 \times 10^{-8} \quad (65)$$

$$\frac{\xi}{2\pi} = 2.1221 \times 10^{-5} \quad (66)$$

$$\xi_{\text{eff}} = 6.6667 \times 10^{-5} \quad (67)$$

$$E_0 = 7.348 \text{ MeV (from exp. masses)} \quad (68)$$

$$E_0^{\text{T0}} = 7.398 \text{ MeV (fractally corrected, } E_0/\sqrt{K_{\text{frak}}}) \quad (69)$$

## Lepton energies

$$E_e = 0.511 \text{ MeV} \quad (70)$$

$$E_\mu = 105.658 \text{ MeV} \quad (71)$$

$$E_\tau = 1776.86 \text{ MeV} \quad (72)$$

## Energy ratios

$$\frac{E_\mu}{E_e} = 206.768 \quad (73)$$

$$\frac{E_\tau}{E_e} = 3477.2 \quad (74)$$

$$\frac{E_\tau}{E_\mu} = 16.817 \quad (75)$$

## .19 Calculation Examples

### Fine structure constant

Given:

- $\xi = 1.3333 \times 10^{-4}$
- $E_0 = 7.398 \text{ MeV}$

Calculation:

$$\left( \frac{E_0}{1 \text{ MeV}} \right)^2 = (7.398)^2 = 54.73 \quad (76)$$

$$\alpha = 1.3333 \times 10^{-4} \times 54.73 \quad (77)$$

$$= 7.297 \times 10^{-3} \quad (78)$$

$$= \frac{1}{137.04} \quad (79)$$

Experimental:  $\alpha_{\text{exp}} = \frac{1}{137.036}$

Deviation: 0.0005 % (corrected on 30 Sept. 2026: previously 0.03 %)

## Characteristic length (electron)

Given:

- $E_e = 0.511 \text{ MeV} = 0.511 \times 1.6 \times 10^{-13} \text{ J} = 8.2 \times 10^{-14} \text{ J}$
- $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
- $c = 3 \times 10^8 \text{ m/s}$

Conversion to SI units:

$$r_0 = \frac{2GE}{c^4} \approx 1.35 \times 10^{-57} \text{ m} \quad (80)$$

Planck comparison:

$$\frac{r_0}{\ell_P} = \frac{1.35 \times 10^{-57}}{1.6 \times 10^{-35}} \approx 8.4 \times 10^{-23} \quad (81)$$

(Corrected on 29 Sept. 2026: previously  $r_0 = 2GE \approx 10^{-28} \text{ m}$  and  $r_0/\ell_P \approx 10^7$  – with the SI values given, the formula reads  $r_0 = 2GE/c^4$  and yields  $1.35 \times 10^{-57} \text{ m}$ ;  $r_0$  thus lies far below the Planck length.)

## 0.1 Symbol Reference

| Symbol               | Meaning                             | Dimension    |
|----------------------|-------------------------------------|--------------|
| $\xi$                | Fundamental parameter               | 1            |
| $E_0$                | Characteristic energy               | $E$          |
| $E_{\text{field}}$   | Universal energy field              | $E$          |
| $T_{\text{field}}$   | Intrinsic time field                | $E^{-1}$     |
| $r_0$                | T0 characteristic length            | $L = E^{-1}$ |
| $t_0$                | T0 characteristic time              | $T = E^{-1}$ |
| $\ell_P$             | Planck length                       | $L = E^{-1}$ |
| $G$                  | Gravitational constant              | $E^{-2}$     |
| $\alpha$             | Fine structure constant             | 1            |
| $E_e, E_\mu, E_\tau$ | Lepton energies                     | $E$          |
| $m_e, m_\mu, m_\tau$ | Lepton masses (= $E$ in nat. units) | $E$          |
| $\mathcal{L}$        | Lagrangian density                  | $E^4$        |
| $\square$            | d'Alembert operator                 | $E^2$        |
| $\xi_{\text{eff}}$   | Effective parameter ( $\xi/2$ )     | 1            |

## 0.2 Unit Conversions

Natural  $\rightarrow$  SI

$$1 \text{ (Energy)} = 1 \text{ GeV} = 1.6 \times 10^{-10} \text{ J} \quad (82)$$

$$1 \text{ (Length)} = \frac{\hbar c}{1 \text{ GeV}} = 0.197 \text{ fm} \quad (83)$$

$$1 \text{ (Time)} = \frac{\hbar}{1 \text{ GeV}} = 6.58 \times 10^{-25} \text{ s} \quad (84)$$

### Standard natural units

In standard convention ( $\hbar = c = 1$ ):

- $\alpha = \frac{e^2}{4\pi\epsilon_0} \approx \frac{1}{137}$  (dimensionless)
- All quantities in powers of energy
- Physical predictions identical to other conventions

### 0.3 Relations to Other Documents

- **Document 003:** Time-mass duality, foundations, origin of  $\xi$
- **Document 011:** Fine structure constant (geometric derivation)
- **Document 012:** Gravitational constant (systematic derivation)
- **Document 018:** Geometric g-2 formulation (fractal geometry)
- **Document 019:** Lagrangian formulation (quantum field theory)
- **Document 026:** Cosmology ( $\xi_{\text{eff}} = \xi/2$ )  
All formulations are based on  $\xi = \frac{4}{3} \times 10^{-4}$ .